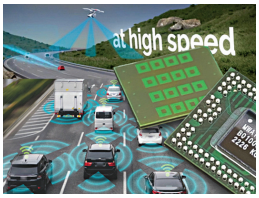
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

First 77GHz beamforming chip for autonomous vehicles

The SPEKTRA from Metawave is the first 77GHz beamforming chip in the front-end antenna in package (AiP) module. The AiP module can enable



accurate, long-distance detection of vehicles beyond 300 metres and pedestrians detection beyond 200 metres. The beamforming chip offers high resolution and a large field of view in a small package. Moreover, the SPEKTRA can be used in all weather conditions, making it suitable for use in autonomous ground and aerial mobility. It integrates analogue and digital MIMO signal processing, which allows distinguishing objects that are close to each other even in the difficult driving scenario and requires lower processing power than all-digital radar and lidars, thus enabling the radar processing to be done at the edge.

Metawave
https://www.metawave.com

Smallest image sensor for eye and face tracking

OG0TB from Omnivision is the world's smallest image sensor launched for face and eye tracking. The ready-to-go image sensor features a 3-layer BSI global shutter sensor that can enable designers to develop more compact and power-efficient devices for use in AR/VR/MR and Metaverse applications. The sensor comes in a compact package measuring just 1.64mm × 1.64mm and features a 2.2µm pixel in a 1/14.46-inch optical format. The sensor consumes less

than 7.2mW at 30 frames per second. For connectivity, the OG0TB supports flexible interfaces, including MIPI with multi-drop, CPHY, SPI, etc. The OmniVision sensor is suitable for designing compact, lightweight, battery-efficient devices such as smart glasses.

Omnivision
https://www.ovt.com

Smallest DSP for 100Gbps coherent transmission

The Steelerton DSP from II-VI Inc. is a digital signal processor (DSP) which enables 100Gbps coherent transmission in optical access networks. It is the industry's first DSP in a pluggable

QSFP28 form factor. The Steelerton DSP is less than one-fifth the size and consumes less than half the power of any other 100Gbps coherent DSP and has an ultra-low power dissipation of 2W. It will enable designers to make a low-cost, power-efficient, and compact transceiver for optical access networks. This innovative module can enable wide-scale deployment of full C-band tunable coherent transceivers into QSFP28 ports.

II-VI Incorporated
https://ii-vi.com

First purpose-built MLSoC for computer vision applications

The MLSoC chip from SiMa.ai is a purpose-built machine learning (ML) system on chip (SoC) suitable for use in a computer vision application. It is the industry's first purpose-built SoC for ML application. The software-centric MLSoC platform can enable quick ML integration

for embedded edge applications. The MLSoC is a power-efficient SoC which delivers the most efficient frames per second/watt. The SoC offers 50 TOPS in a 5W envelope and comes with a dedicated computer vision coprocessor. The MLSoC platform is suitable for use in a wide variety of applications, including robotics, smart vision, autonomous vehicles, drones, etc.

SiMa.ai
https://sima.ai

AI processor for intelligent applications at the edge

The Kinara Ara-1 Edge AI processor delivers a balance between computing performance and power efficiency for intelligent applications at the edge. The Ara-1 Edge processor is suitable for applications that require flexibility and scalability and demand low latency, such as cameras and multi-chip edge servers. The Ara

Edge AI processors can effectively run multiple models without incurring any switch-time performance penalties, gen-